SOLUTIONS LEVEL 1

NCERT / BOARD PATTERN – 150 QUESTIONS

Coverage: Types of solutions, concentration terms, solubility, Henry's law, Raoult's law (ideal/non-ideal), vapour pressure, colligative properties, abnormal molar mass

SECTION 1: BASIC DEFINITIONS & TYPES (Q1–Q20)

1. Define solution in chemistry.
2. What is a binary solution? Give one example.
3. Define solute and solvent with an example from liquid–liquid solution.
4. What is meant by homogeneous mixture? How is it related to a solution?
5. Classify the following as solid, liquid, or gas solutions:
   (a) Air
   (b) Brass
   (c) Sugar in water
6. What are solid solutions? Give two examples.
7. What are liquid solutions? Give two examples.
8. What are gaseous solutions? Give two examples.
9. Define "aqueous solution" and "non-aqueous solution" with examples.
10. What is meant by "saturated solution" at a given temperature?
11. Define unsaturated solution.
12. Define supersaturated solution.
13. What is meant by "ideal solution"?
14. What is a "non-ideal solution"?
15. State two characteristics of an ideal solution.
16. State two characteristics of a non-ideal solution.
17. Explain the term "miscible liquids" with an example.
18. Explain the term "immiscible liquids" with an example.
19. Name the type of solution formed when a gas is dissolved in a solid. Give one example.

20. Name the type of solution formed when a solid is dissolved in another solid. Give one example.

SECTION 2: CONCENTRATION TERMS – MASS %, MOLE FRACTION, MOLARITY, MOLALITY (Q21–Q55)

21. Define mass percentage (% w/w) of a component in a solution.
22. Define volume percentage (% v/v) of a component in a solution.
23. Define mass by volume percentage (% w/v).
24. Define mole fraction of a component in a solution.
25. Define molarity of a solution. Give its unit.
26. Define molality of a solution. Give its unit.
27. What is the SI unit of mole fraction?
28. Which of the following is temperature-dependent: molarity or molality? Explain briefly.
29. Express 10% (w/w) NaCl solution in words.
30. A solution contains 10 g of urea in 90 g of water. Calculate mass percentage of urea.
31. Calculate the mole fraction of water in a solution containing 18 g of glucose ($C_6H_{12}O_6$) in 180 g of water.
32. 0.5 mol of solute is dissolved in 500 g of water. Calculate molality of the solution.
33. 0.2 mol of NaCl is dissolved in enough water to make 1 L of solution. Calculate molarity.
34. Differentiate between molarity and molality (any two points).
35. Why does molarity of a solution change with temperature, but molality does not?
36. A solution contains 2 mol of solute A and 3 mol of solvent B. Calculate mole fraction of A.
37. 5 mol of ethanol and 15 mol of water are mixed. Calculate mole fraction of ethanol.
38. State the relationship between mole fractions of solute and solvent in a binary solution.
39. How many moles of solute are present in 250 mL of 0.2 M solution?
40. How many grams of NaOH are present in 500 mL of 0.1 M NaOH solution? (Molar mass NaOH = 40 g mol^{-1}.)
41. Define "parts per million" (ppm). In which type of solutions is it commonly used?

42. Express 0.9% (w/v) saline solution in g of solute per 100 mL of solution.
43. A 1 molal solution of glucose in water contains what mass of solute and solvent? (Assume 1 kg of solvent.)
44. How is normality (N) defined? In which type of reactions is it often used?
45. For a given solution, what happens to molarity if temperature is increased?
46. For a given solution, what happens to molality if temperature is increased?
47. In highly dilute aqueous solutions, how are molarity and molality approximately related?
48. A solution is prepared by dissolving 4 g NaOH in enough water to make 500 mL of solution. Calculate its molarity.
49. What volume of 2 M HCl is needed to prepare 500 mL of 0.5 M HCl solution?
50. Explain the term "stock solution" in context of molarity.
51. Why is mole fraction often preferred in vapour-pressure and colligative property calculations?
52. What is meant by "mole percent"?
53. A solution contains 20% (w/w) ethanol and has density 0.95 g mL^{-1}. Explain how you would convert this to molarity (no calculation required).
54. What is meant by "dilution" of a solution? Write the formula relating M_1, V_1, M_2, V_2.
55. State one advantage and one disadvantage of expressing concentration as mass percentage.

SECTION 3: SOLUBILITY, FACTORS & HENRY'S LAW (Q56–Q85)

56. Define solubility of a solute in a solvent.
57. How is solubility of a solid in liquid expressed (units)?
58. State any two factors affecting solubility of a solid in a liquid.
59. Explain qualitatively how temperature affects solubility of most solids in water.
60. How does solubility of a gas in a liquid vary with pressure?
61. State Henry's law for solubility of a gas in a liquid.
62. Write the mathematical expression for Henry's law and explain the terms.
63. What are Henry's law constant (K_H) and its significance?

64. According to Henry's law, what happens to solubility of a gas if pressure is doubled at constant temperature?
65. How does temperature affect solubility of gases in liquids?
66. Why does aerated water taste better when cooled?
67. Why do deep-sea divers face the problem of "bends"? Relate it to Henry's law.
68. Why is it dangerous to open a soda bottle immediately after shaking?
69. Why do fish die in water contaminated with hot effluents from factories?
70. Give one application of Henry's law in the packaging of soft drinks.
71. Give one application of Henry's law in deep-sea diving (helium–oxygen mixture).
72. For a particular gas–solvent system, K_H is large. What does it indicate about the solubility of the gas?
73. For a particular gas, K_H is smaller at lower temperature. What does it tell about solubility?
74. Write one limitation of Henry's law.
75. When does Henry's law deviate significantly from ideal behaviour? Name any one condition.
76. Explain why oxygen is less soluble in water than CO_2, although both obey Henry's law approximately.
77. How does nature of the gas affect its solubility in water? Give one example.
78. At a given temperature, solubility of gas A in water is higher than that of gas B. What can you say about their Henry's constants?
79. Why is the solubility of gases in organic solvents often higher than in water?
80. State the units in which Henry's constant is commonly expressed.
81. What is meant by "pressure of a gas over the solution"?
82. Name two colligative properties that can be used to determine solubility of a sparingly soluble solute indirectly.
83. Explain qualitatively how salting out of soap (precipitation by NaCl) relates to solubility.
84. What is meant by "common ion effect" in context of solubility?
85. How does common ion effect influence solubility of sparingly soluble salts?

SECTION 4: IDEAL & NON-IDEAL SOLUTIONS, RAOULT'S LAW (Q86–Q115)

86. State Raoult's law for a solution containing non-volatile solute and volatile solvent.
87. Write Raoult's law expression for vapour pressure of solvent in ideal solution.
88. In a binary ideal solution of volatile liquids A and B, write expression for total vapour pressure.
89. Define ideal solution in terms of Raoult's law.
90. State any two characteristics of an ideal solution regarding enthalpy change and volume change on mixing.
91. Define non-ideal solution.
92. What are positive deviations from Raoult's law?
93. What are negative deviations from Raoult's law?
94. Give one example each of solutions showing positive and negative deviation from Raoult's law.
95. What is azeotrope?
96. Distinguish between minimum boiling azeotrope and maximum boiling azeotrope.
97. Give one example each of minimum and maximum boiling azeotrope.
98. For an ideal solution, what is ΔH_{mix} and ΔV_{mix}?
99. For a solution showing positive deviation, what is the sign of ΔH_{mix} and ΔV_{mix}?
100. For a solution showing negative deviation, what is the sign of ΔH_{mix} and ΔV_{mix}?
101. On a vapour-pressure vs composition curve, how do you recognize positive deviation?
102. On the same graph, how do you recognize negative deviation?
103. Name the forces responsible for negative deviation from Raoult's law.
104. Name the type of intermolecular forces that lead to positive deviation from Raoult's law.
105. What is meant by "partial vapour pressure" of a component in a solution?
106. Write expression for partial vapour pressure of component A in an ideal binary solution according to Raoult's law.
107. How is mole fraction of a component in vapour phase related to partial pressures (Dalton's law)?
108. State Dalton's law of partial pressures.

109. Write relation between total vapour pressure and mole fraction of components in an ideal binary solution.

110. Why is Raoult's law considered a special case of Henry's law?

111. Explain briefly: all gases obey Henry's law approximately, but only some liquid mixtures obey Raoult's law ideally.

112. Under what condition can a non-ideal solution become nearly ideal?

113. Why are mixtures of chemically similar liquids (like benzene–toluene) closer to ideal behaviour?

114. Why does ethanol–water mixture show positive deviation from Raoult's law?

115. Why does acetone–chloroform mixture show negative deviation from Raoult's law?

SECTION 5: COLLIGATIVE PROPERTIES (INTRO + QUALITATIVE) (Q116–Q150)

116. What are colligative properties? Give two examples.

117. Why are they called "colligative"?

118. Name four colligative properties of solutions.

119. How is relative lowering of vapour pressure defined?

120. Define elevation in boiling point.

121. Define depression in freezing point.

122. Define osmotic pressure.

123. State the relation between lowering of vapour pressure and mole fraction of non-volatile solute.

124. Why is boiling point of solution higher than that of pure solvent?

125. Why is freezing point of solution lower than that of pure solvent?

126. Write the formula relating elevation in boiling point ΔT_b with molality m.

127. What is molal elevation constant K_b? On what does it depend?

128. Write the formula relating depression in freezing point ΔT_f with molality m.

129. What is molal depression constant K_f? On what does it depend?

130. What is the van't Hoff factor i?

131. How is van't Hoff factor defined in terms of observed and calculated colligative property?

132. Why do electrolytes show abnormal colligative properties?

133. Why do non-electrolytes usually obey colligative property equations accurately?

134. Write van't Hoff equation for osmotic pressure π of a dilute solution.

135. What is the unit of osmotic pressure in SI system?

136. State one advantage of osmotic pressure method over other colligative properties for molar mass determination.

137. Why is hypertonic solution harmful to red blood cells?

138. What is hypotonic solution? How does it affect a cell?

139. What is isotonic solution? Give one medical application.

140. How would you detect association of solute molecules (e.g., acetic acid in benzene) using colligative properties?

141. How would you detect dissociation of solute molecules (e.g., NaCl in water) using colligative properties?

142. A solution shows higher depression in freezing point than expected. What does it indicate about solute behaviour?

143. How does presence of a non-volatile solute affect vapour pressure of solution?

144. Define relative lowering of vapour pressure and give its expression in terms of moles of solute and solvent.

145. Why is elevation in boiling point a very small quantity for dilute solutions?

146. Name any two experimental methods used to measure colligative properties.

147. How can colligative properties be used to calculate degree of dissociation (α) of an electrolyte?

148. Why are colligative properties valid only for dilute solutions?

149. State the conditions under which Raoult's law and colligative property relations hold best.

150. A solution of glucose and urea in water gives the same depression in freezing point. What can be said about their total number of moles in solution?